

Problem 1: Put a penny on each gray square of strip A. A jump: a peg hops over a peg right next to it, in a straight line, into the empty square just beyond, and the peg it hopped over comes off the strip. Make one jump. Draw an X in the square where the peg lands.

Strip A

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Problem 2: On strips B, C and D, put pennies on the gray squares and try to make a jump. Circle *possible* or *not possible* for each strip. On strip D, draw an X in every square a peg could land on with one jump.

Strip B

circle one: possible not possible

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Strip C

circle one: possible not possible

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Strip D

circle one: possible not possible

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Problem 3: On each of strips E to H, put pennies on the gray squares. Jump until no jump is possible, in either direction. Write the number of pegs left. Then set the strip up again and jump in a different order. Write the fewest pegs you managed to leave.

Strip E

pegs left, first try: _____ fewest: _____

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Strip F

pegs left, first try: _____ fewest: _____

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Problem 3, continued: for strips G and H, put pennies on the gray squares, jump until no jump is possible and write the number of pegs left; then set up again, jump in a different order, and write the fewest.

Strip G

pegs left, first try: _____ fewest: _____

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Strip H

pegs left, first try: _____ fewest: _____

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Problem 4: Put five pennies on strip I in a pattern that you think can be cleared down to one peg. Play it. If it gets stuck with more than one peg, set up a new pattern on strip J and play that. When you find a pattern that works, draw a dot in each of its five squares. Then use strips K and L the same way to find a pattern of five pegs that always gets stuck with three or more pegs left.

Strip I

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Strip J

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Strip K

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Strip L

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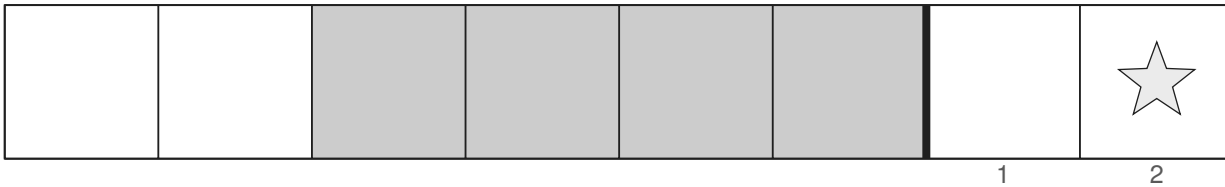
Problem 5: On strip M the thick line is the river, and the squares to the left of it are the home side. Put pennies on the two gray squares. Cross the river with one jump. Draw an X where the peg lands. Strips N, O, P and Q start with three, four, four and six pegs on the home side. For each strip, put pennies on the gray squares, jump as many times as you like, and try to get a peg onto square 2, the square with the star. Write the farthest square any peg reached: 0, 1 or 2, where 0 means that no peg crossed the river.

Strip M**Strip N**

farthest square reached (0, 1 or 2): _____

**Strip O**

farthest square reached (0, 1 or 2): _____

**Strip P**

farthest square reached (0, 1 or 2): _____

**Strip Q**

farthest square reached (0, 1 or 2): _____



Problem 6: On strips R1, R2 and R3, choose your own starting squares on the home side, up to six pennies, a different set on each strip. Draw a dot in each square you used. Then play and write the farthest square reached, 0, 1 or 2.

Strip R1

farthest square reached (0, 1 or 2): _____

								
							1	2

Strip R2

farthest square reached (0, 1 or 2): _____

								
							1	2

Strip R3

farthest square reached (0, 1 or 2): _____

								
							1	2

Problem 7: Strip S has a value printed in the corner of each square. The total of a position is the sum of the values of the squares that have pegs on them. For each row of the table: put pennies on the squares listed, write the total, make the jump described, and write the new total. In the last column write *up*, *same* or *down*.

Strip S

1	1	2	3	5	8	13	21	
							1	2

pegs on the squares valued ...	total before	total after	up, same or down
all six home squares; make the only jump that is possible	_____	_____	_____
2 and 3; jump to the left	_____	_____	_____
2 and 3; jump to the right	_____	_____	_____
5 and 8; jump to the right, across the river	_____	_____	_____
8 and 13, one on each side of the river; jump to the left	_____	_____	_____

Problem 8: Write the total of the six home squares of strip S: _____ and the value of square 2: _____. Can a peg ever land on square 2 of strip S, whatever jumps you make? Use the last column of your table to tell the adult at your table how you know.

Problem 9: Strip T has seven home squares and no values. Write 1 and 1 in the corners of the first two home squares, then continue to the right: each value is the sum of the two values before it, up to the star. Write the total of the seven home squares: _____ and the value of square 2: _____. Can a peg reach square 2 of strip T? Say how you know.

Strip T

									
								1	2

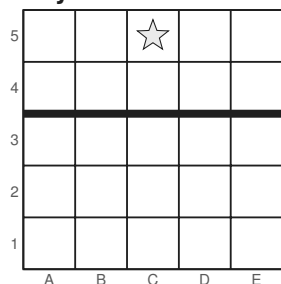
Problem 10: Board U is a grid with the river across it. Squares are named by column letter and row number, like C3. On a board, jumps go up, down, left or right, never diagonally. Put pennies on C3 and C2, make the one jump that crosses the river, and write the name of the square where the peg lands:

Board U

5					
4					
3					
2					
1					
	A	B	C	D	E

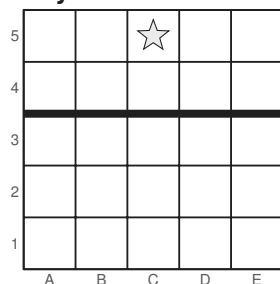
Problem 11: Use board U again (on page 5). Put only three pennies on home squares of your choice and try to get a peg to the star at C5. Try at least four different placements. Record each one on a small board below: draw a dot in each square you used and write the highest row number any peg reached.

Try 1



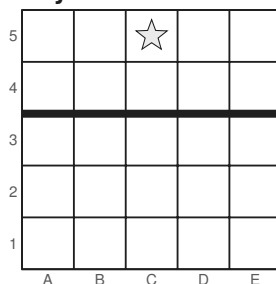
highest row: _____

Try 2



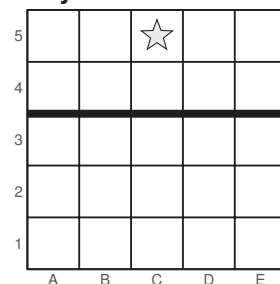
highest row: _____

Try 3



highest row: _____

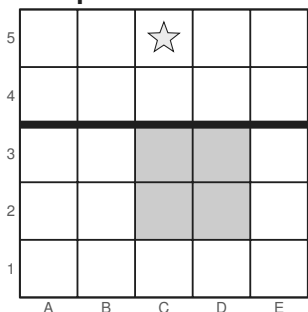
Try 4



highest row: _____

Problem 12: Each small picture shows a shape of four pegs on the home side of board U. For each shape, copy it onto board U (on page 5) with pennies and try to get a peg to the star at C5, jumping as many times as you like. Under the picture write *yes* if a peg reached the star and *no* if it never did. For each shape that works, also write its jumps in order, like C1 over C2 to C3.

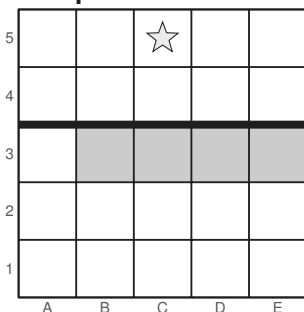
Shape 1



works: _____

jumps: _____

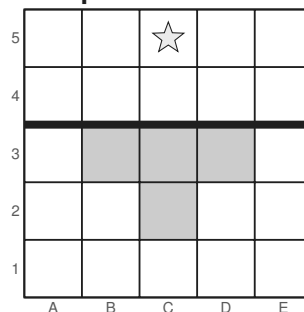
Shape 2



works: _____

jumps: _____

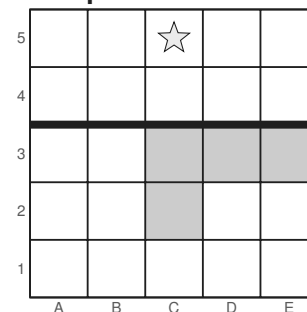
Shape 3



works: _____

jumps: _____

Shape 4



works: _____

jumps: _____

Problem 13: Board V shows a value in the corner of every square of board U. For each line, put pennies on the squares named, make the jump, and write the total before and the total after.

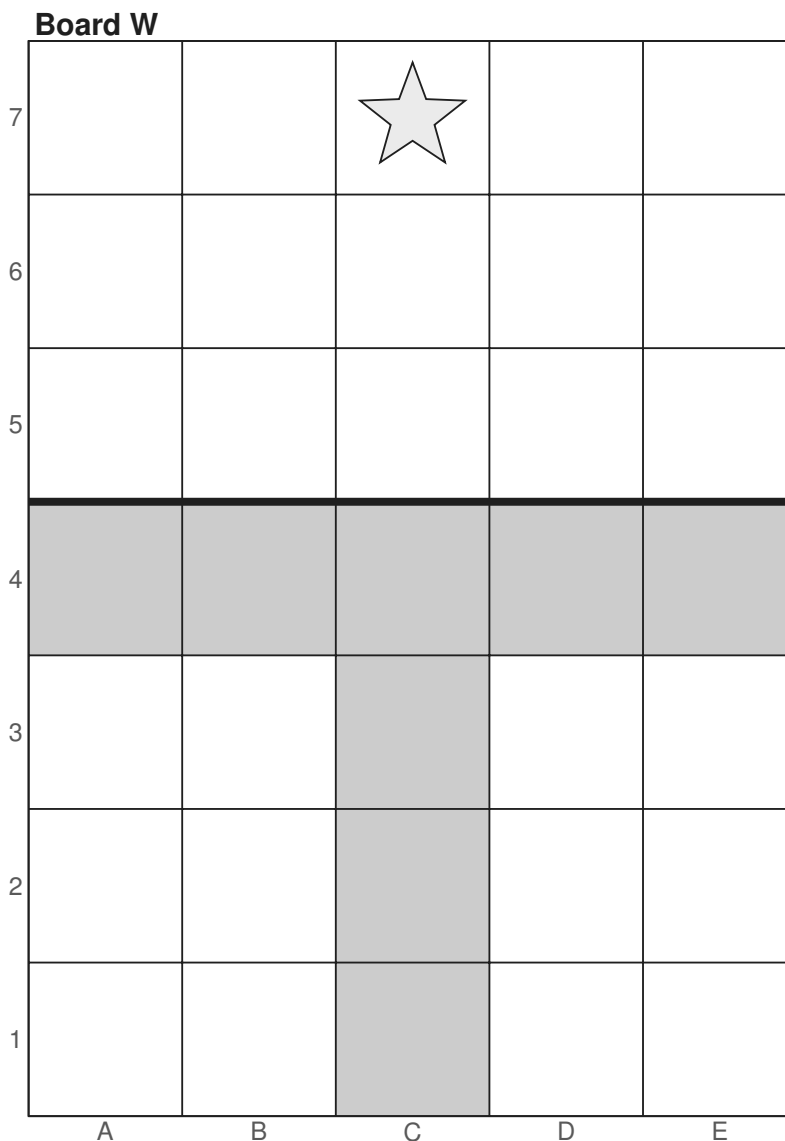
C3 and C2; jump up	before _____	after _____
D3 and E3; jump to the left	before _____	after _____
B3 and C3; jump to the right	before _____	after _____
C4 and C3; jump down	before _____	after _____

Board V

5	8	13	21	13	8
4	5	8	13	8	5
3	3	5	8	5	3
2	2	3	5	3	2
1	1	2	3	2	1
	A	B	C	D	E

Problem 14: On board V, find the three home squares with the largest values and write their total: _____. Write the value of the star: _____. Then write the total of the four squares of shape 1: _____ and of shape 4: _____. Tell the adult at your table what these totals say about Problem 11, and what shape 1 shows.

Problem 15: Board W has eight pegs: five along the river and three below the middle one. Put pennies on the gray squares. Find seven jumps that put a peg on the star at C7. Write down each jump as you make it: the square the peg jumps from, the square it jumps over, and the square it lands on. If you get stuck, set the pegs up again and write a fresh list.



Jump 1: from _____ over _____ to _____

Jump 5: from _____ over _____ to _____

Jump 2: from _____ over _____ to _____

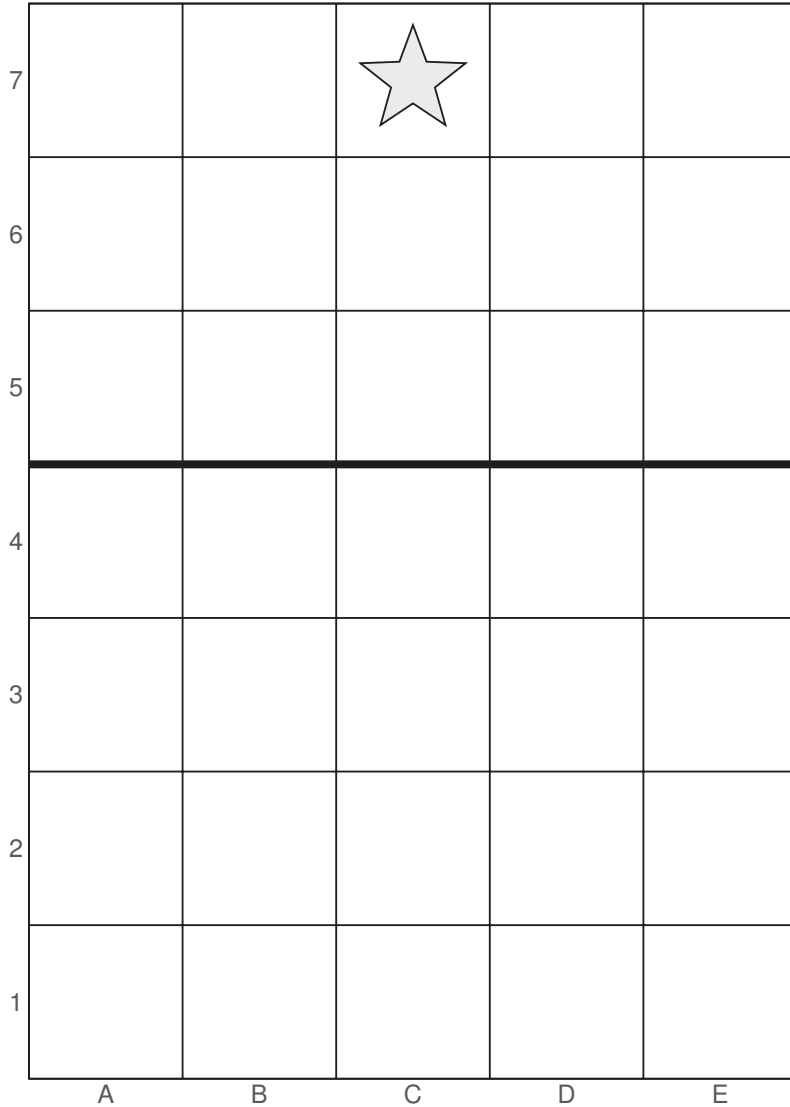
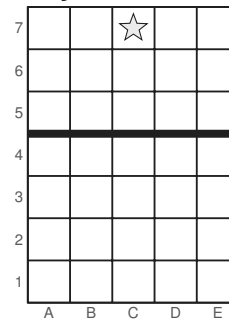
Jump 6: from _____ over _____ to _____

Jump 3: from _____ over _____ to _____

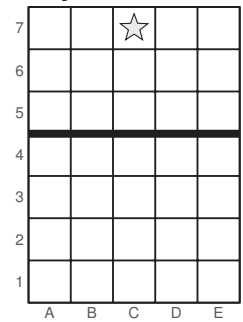
Jump 7: from _____ over _____ to _____

Jump 4: from _____ over _____ to _____

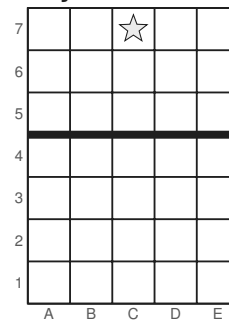
Problem 16: Board X is empty. Find a set of eight home squares, different from the shape of board W, from which a peg can reach the star at C7. Test each set with pennies on board X. Record each set you try on one of the small boards: a dot in each square, and the highest row reached. When a set works, draw a dot in each of its eight squares on board X and list its seven jumps.

Board X**Try 1**

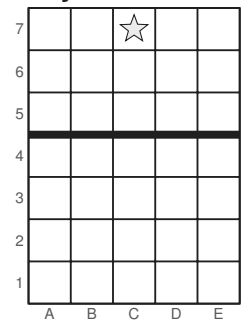
highest row: ____

Try 2

highest row: ____

Try 3

highest row: ____

Try 4

highest row: ____

Jump 1: from ____ over ____ to ____

Jump 5: from ____ over ____ to ____

Jump 2: from ____ over ____ to ____

Jump 6: from ____ over ____ to ____

Jump 3: from ____ over ____ to ____

Jump 7: from ____ over ____ to ____

Jump 4: from ____ over ____ to ____